
Vision-Language Models for Interpretation and Classification of Echocardiographic Views

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Abstract

Echocardiographic view classification is a critical task in cardiac imaging, necessary for accurate diagnosis and treatment planning. While current deep learning methods excel at image-based classification, they often lack the ability to provide reasoning or step-by-step justification for their decisions, limiting their utility in clinical settings. In this work, we propose a novel approach leveraging reinforcement learning and a clinically-informed reward model to align a pre-trained vision-language model. The reward model, designed based on clinical standard guidelines, carefully evaluates the reasoning capabilities of the vision-language model (VLM) and provides scalar feedback for optimization. This alignment enables the VLM to generate detailed reasoning, interpretations, and step-by-step justifications for echocardiographic images. By incorporating clinical reasoning into the decision-making process, our approach enhances classification accuracy and interpretability, contributing to more transparent and effective AI-assisted diagnostics in echocardiography.

1 Introduction

Transthoracic echocardiography (TTE) is a non-invasive imaging technique that utilizes ultrasound waves to visualize the structure and function of the heart. It is widely used for diagnosing a range of cardiac conditions, including valve diseases, heart failure, and congenital heart defects [9]. TTE is crucial in clinical practice as it provides real-time, dynamic imaging, allowing physicians to assess cardiac health without invasive procedures [16].

In echocardiography, multiple 2D views are used to capture specific regions of the heart’s 3D anatomy, with each view offering unique insights. Accurately identifying these views is crucial for making informed diagnostic and therapeutic decisions [6]. As ultrasound technology is increasingly adopted by non-cardiologists across various specialties, often without advanced expertise, the demand for tools that aid in view recognition and in extracting key echocardiographic features has grown, particularly for novice users [16].

A reasoning-driven interpretation of echocardiographic images can provide physicians with valuable insights into complex cardiac anatomy and function. By offering step-by-step justifications and interpretative analyses of key features such as chamber size, valve morphology, and blood flow patterns, reasoning-based explanations can complement visual information, enabling clinicians to better understand the decision-making process behind classifications. Previous work [20] has demonstrated the potential of linking medical image classification with predefined textual concepts. However, these models are limited by their reliance on a narrow set of predefined concepts, restricting their ability to generalize across diverse clinical scenarios. Moreover, they fail to provide cohesive and interpretable reasoning that connects individual concepts into a comprehensive narrative, which

is essential for supporting clinical decision-making and enhancing transparency in AI-assisted diagnostics.

Recent advancements in large language models (LLMs) have revolutionized natural language processing, achieving remarkable performance in tasks such as text generation, question answering, and contextual reasoning [1]. These successes can largely be attributed to their training on vast datasets and fine-tuning through techniques like reinforcement learning with human feedback (RLHF) [10], which aligns their outputs with human preferences and desired behaviors. The effectiveness of such alignment strategies in LLMs has inspired similar efforts in multi-modal models, particularly vision-language models (VLMs), which integrate visual and textual modalities to tackle complex tasks that require both reasoning and interpretability.

Building on these principles, vision-language multi-modal models have demonstrated impressive results in tasks such as image captioning, visual question answering, and zero-shot classification [2, 7, 19]. Despite these achievements, their application to domains like echocardiography, where reasoning and interpretability are critical, remains underexplored. These advancements provide a promising foundation for aligning VLMs to exhibit desired reasoning capabilities and step-by-step justification, making them suitable for specialized tasks such as echocardiographic view classification.

In this work, we propose using reinforcement learning with a proximal policy optimization (PPO) [13] approach to fine-tune a pre-trained VLM. A reward model is designed based on clinical standard guidelines evaluates the reasoning capability of the VLM. The reward model assesses the alignment of the VLM’s reasoning with clinically relevant observations and provides scalar feedback to iteratively update the model. This alignment process enables the VLM to generate detailed anatomical and clinical observations to justify its predicted echocardiographic view. Experimental results show that the VLM exhibits significant improvements in its reasoning and interpretative capabilities. Specifically, the VLM can now generate detailed explanations and step-by-step justifications of its predictions, including key anatomical and clinical features of the input. While not entirely accurate, these explanations represent a meaningful step toward more interpretable and clinically relevant AI systems for echocardiography.

2 Background

2.1 Incorporating Reasoning into LLMs via GRPO

Recent advancements in reinforcement learning (RL) have introduced the Group Relative Policy Optimization (GRPO) algorithm, a novel approach designed to enhance the reasoning capabilities of Large Language Models (LLMs) without relying on supervised fine-tuning. GRPO was first introduced in the DeepSeek-R1 framework [14], demonstrating significant improvements in logical inference and mathematical reasoning tasks.

Building upon this foundation, the VLM-R1 framework [15] extends GRPO to Vision-Language Models, enabling them to perform complex visual reasoning tasks. VLM-R1 showcases the effectiveness of GRPO in multimodal settings, surpassing traditional supervised fine-tuning methods in both performance and generalization. Further exploration into visual perception tasks led to the development of Visual-RFT [8], which applies GRPO to enhance visual understanding capabilities. This approach emphasizes the adaptability of GRPO across various domains within computer vision.

In the medical field, the integration of GRPO has shown promising results. MedVLM-R1 [11] employs GRPO to incentivize medical reasoning in Vision-Language Models, particularly in radiological image analysis. Similarly, Med-R1 [5] leverages GRPO to improve generalizable medical reasoning across diverse imaging modalities, including CT, MRI, and ultrasound. These developments underscore the versatility and effectiveness of GRPO in fostering reasoning abilities across different model architectures and application domains.

2.2 Medical Images View Classification

Recent advancements in deep learning have significantly improved echocardiographic view classification, a critical task for cardiac diagnosis and function assessment. Early work by [9] used a CNN-based approach inspired by VGG-16, achieving over 97% accuracy across 15 echocardiographic views while surpassing expert performance in some cases. Building on this, [12] leveraged transfer

learning to fine-tune pre-trained convolutional networks, addressing data scarcity challenges and enhancing view-specific feature extraction for improved classification. More recently, [6] proposed a multi-task framework that integrates view classification with real-time image quality assessment, optimizing both diagnostic reliability and workflow efficiency. Collectively, these studies highlight the growing importance of automated view classification in reducing clinician workload, enhancing diagnostic accuracy, and improving patient outcomes.

3 Methodology

3.1 Review of GRPO

Group Relative Policy Optimization (GRPO) is a reinforcement learning algorithm proposed to enhance the reasoning capabilities of large language models by leveraging group-level relative performance during policy updates. Originally introduced within the DeepSeek framework [17], GRPO differs significantly from traditional reinforcement learning approaches by eliminating the need for explicit value function approximation. Instead, GRPO directly estimates a baseline reward from the sampled group responses, thereby reducing computational overhead and promoting stable convergence.

Formally, for a given input prompt x , GRPO generates a set of candidate responses from the current policy $\pi_\theta(y|x)$. Each response y_i within this group is evaluated based on a reward function $r(x, y_i)$, reflecting its performance relative to the desired criteria, such as accuracy or logical consistency. The group average reward $\mu(x)$ serves as a baseline to compute the relative advantage $A(x, y_i) = r(x, y_i) - \mu(x)$. The policy is then updated to preferentially increase the likelihood of responses exhibiting positive advantages and decrease the likelihood of those with negative advantages, effectively guiding the model toward superior reasoning outcomes.

The optimization objective of GRPO is formulated as:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{G} \sum_{i=1}^G \frac{\pi_\theta(y_i|x)}{\pi_{\text{old}}(y_i|x)} A(x, y_i) - \beta D_{\text{KL}}(\pi_\theta(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)) \right], \quad (1)$$

where π_θ denotes the updated policy, π_{old} represents the policy used to sample the group responses, and π_{ref} is a reference policy typically derived from supervised fine-tuning. Additionally, β is a hyperparameter that regulates the strength of the Kullback–Leibler divergence regularization term, ensuring policy updates remain close to the established reference, thereby maintaining stability during optimization.

In the DeepSeek-R1 framework, GRPO is employed with two primary rule-based reward functions: an accuracy reward and a format reward. The accuracy reward quantifies the correctness of the final answer with respect to ground-truth labels, such as validating a mathematical result or confirming that generated code passes designated unit tests. In contrast, the format reward enforces output consistency by requiring the model to adopt a structured format in its responses, commonly involving reasoning annotations encapsulated in template-like tokens (e.g., ...). Together, these rewards facilitate both precise and interpretable model outputs, fostering reliable reasoning behavior in the fine-tuned policy [14].

3.2 Our Method

In this subsection, we outline our approach, which consists of three main steps, including 1) *View Classification Fine-Tuning*, 2) *Reward Model Design*, and 3) *RL Fine-tuning*. This framework is illustrated in 3.2.

View Classification Fine-Tuning Step. We begin by fine-tuning the VLM in a supervised manner, where the model is trained to predict a single token corresponding to the view classification. The model receives as input both the text prompt t and the corresponding image v , with the goal of outputting a token that represents the predicted view class. The new policy for this task is denoted as $\pi_\theta(y|t, v)$, where y is the predicted view, and θ represents the parameters of the model. The objective of this step is to minimize the cross-entropy loss between the predicted view token y and the true label y_{gt} .

This step is essential, as it enables the model to learn the visual features and representations necessary to differentiate between different echocardiographic views. The model becomes capable of associating specific visual characteristics of the images with predefined view classes, which is fundamental for the subsequent reinforcement learning stages.

Design of a Soft Clinical Reward Function Existing reward mechanisms for view classification consider only the final predicted label when computing reinforcement rewards. While this approach can yield high classification accuracy, it allows the model to produce incorrect or clinically unsound reasoning chains that happen to lead to a correct answer. In the context of echocardiography view classification, such spurious reasoning is unacceptable, since clinical decisions rely on both the correctness of the view label and the validity of the underlying anatomical justification.

To address this, we propose a soft clinical reward function that awards partial credit based on the quality of the generated reasoning. We first extract, from standard clinical guidelines, key anatomical landmarks, structural features, and orientation descriptors for each echo view. These criteria form positive and negative term sets which we use to evaluate the model’s reasoning text alongside its predicted label.

Let $\mathcal{C} = \{c_1, \dots, c_M\}$ be the set of M echo-view classes, and let K denote the number of clinical criterion types (e.g., landmarks, features, orientation). For each class $c \in \mathcal{C}$ and criterion $k \in \{1, \dots, K\}$, define the positive term set

$$P_c^{(k)} = \{\tau_{c,1}^{(k)}, \dots, \tau_{c,|P_c^{(k)}|}^{(k)}\},$$

and the negative term set

$$N_c^{(k)} = \bigcup_{c' \in \mathcal{C} \setminus \{c\}} P_{c'}^{(k)}.$$

Given a model prediction $\hat{c}$ and its reasoning text τ , we compute:

1. Answer reward:

$$R_{\text{ans}}(\hat{c}, c) = \mathbf{1}\{\hat{c} = c\}.$$

2. Reasoning-based reward: For each k , let

$$S^{(k)}(\hat{c}, \tau) = \frac{1}{|P_{\hat{c}}^{(k)}|} \sum_{t \in P_{\hat{c}}^{(k)}} \mathbf{1}\{t \in \tau\}, \quad U^{(k)}(\hat{c}, \tau) = \frac{1}{|N_{\hat{c}}^{(k)}|} \sum_{t \in N_{\hat{c}}^{(k)}} \mathbf{1}\{t \in \tau\}.$$

Aggregating with nonnegative weights $\{w_k\}$ such that $\sum_k w_k = 1$ and penalty scale $\alpha > 0$ yields

$$R_{\text{reason}}(\hat{c}, \tau) = \max\left(0, \min\left(1, \sum_{k=1}^K w_k S^{(k)}(\hat{c}, \tau) - \alpha \sum_{k=1}^K w_k U^{(k)}(\hat{c}, \tau)\right)\right).$$

3. Final soft reward:

$$r(c, \hat{c}, \tau) = \gamma R_{\text{ans}}(\hat{c}, c) + (1 - \gamma) R_{\text{reason}}(\hat{c}, \tau).$$

In our implementation we set $K = 3$ (landmarks, features, orientation), $(w_1, w_2, w_3) = (0.4, 0.4, 0.2)$, $\alpha = 0.5$, and $\gamma = 0.6$. This formulation extends naturally to any number of classes and clinical criteria, ensuring that the model is rewarded not only for correct view predictions but also for clinically sound reasoning aligned with established guidelines.

RL Fine-Tuning In the final training stage, we replace conventional PPO-based updates with Group Relative Policy Optimization (GRPO) [14] to directly optimize our policy against the soft clinical reward. To provide the model with essential anatomical context while avoiding explicit label hints, we prepend a set of structural clues to the text prompt. These clues enumerate key landmarks, salient features, and preferred orientations derived from clinical guidelines, but do not reveal which echo view is correct. During each update, the policy samples a small group of reasoning-and-answer pairs for the same image–prompt tuple and computes each sample’s reward according to the formulation in Section 3.2. By centering these rewards within the group, GRPO emphasizes relative improvements:

samples whose combined view prediction and reasoning exceed the group average receive positive advantage, whereas less coherent outputs are down-weighted. This mechanism naturally encourages the model to learn mappings between structural clues, image content, and the correct anatomical explanation without mode collapse.

Our policy network $\pi_{\theta}(y | t, v)$ remains conditioned on both the augmented text prompt t (original instruction plus structural clues) and the echocardiographic image v . At initialization, π_{θ} is warm-started from the supervised fine-tuned model to ensure basic competency. GRPO updates then adjust the conditional distribution over full reasoning traces and final answers, promoting those that yield higher soft clinical rewards. To preserve stability and prevent excessive deviation from the initial policy, we retain a KL-divergence barrier anchored to the reference policy. This combination of structural prompting and group-based advantage estimation allows the model to acquire clinically aligned reasoning pathways in addition to accurate view classification.

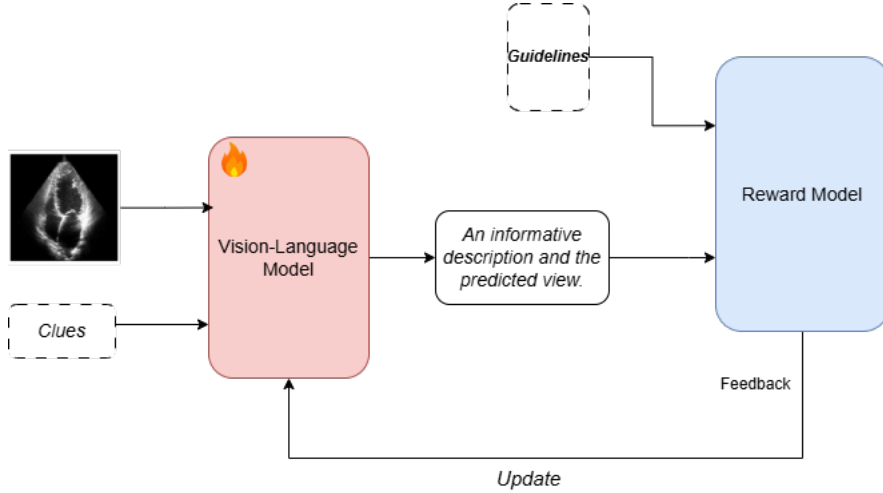


Figure 1: Illustration of the proposed Vision-Language Model (VLM) framework for echocardiogram view classification. The VLM takes an echo image and textual "clues" as input, generating view predictions. These are evaluated by a rule-based reward model, providing feedback that fine-tunes the VLM through reinforcement learning.

4 Experiments

4.1 Dataset

The dataset used in this project was obtained from the echocardiography database at Vancouver General Hospital, following approval from the Clinical Medical Research Ethics Board and in consultation with the Information Privacy Office. The data includes 3,151 unique patients diagnosed with various heart conditions between 2011 and 2015. In total, the dataset comprises 16,612 echocardiogram cine loops (amounting to 807,908 frames) captured from 12 standard views. The examples of images from each class shown in Fig. 4.1. For the entire dataset, a senior cardiologist manually labeled each cine series with its corresponding view class.

4.2 Setting

Training was conducted using an 8×NVIDIA V100 GPU setup, each equipped with 32 GB of memory, to manage the computational load of fine-tuning the Vision-Language Model (VLM). The pre-trained model used for this task was Qwen2-VL-2B-Instruct [18], an open-source model known for its efficiency in terms of parameter size compared to other state-of-the-art VLMs, which often contain significantly more parameters. This efficiency made it feasible to fine-tune the model within the resource constraints of the available hardware. Given the limited resources, LoRA [4] (Low-Rank Adaptation) was employed to perform efficient fine-tuning. LoRA adapts large models by adding

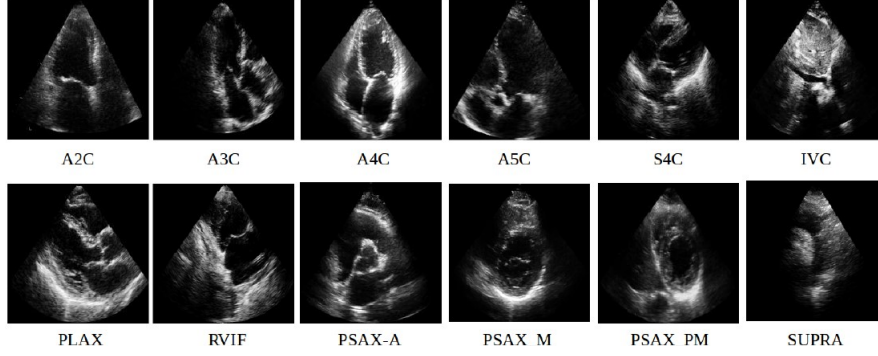


Figure 2: Example of echocardiography images.

trainable low-rank matrices to pre-trained weights, enabling effective training with fewer updated parameters and optimizing both memory usage and computational throughput.

Due to resource constraints, we set the GRPO group size to 3, which is relatively small and can lead to high variance; results could potentially improve with a larger group size.

For the RL fine-tuning phase, we fine-tuned the entire model for one epoch. The corresponding training curves are shown in Figure ??.

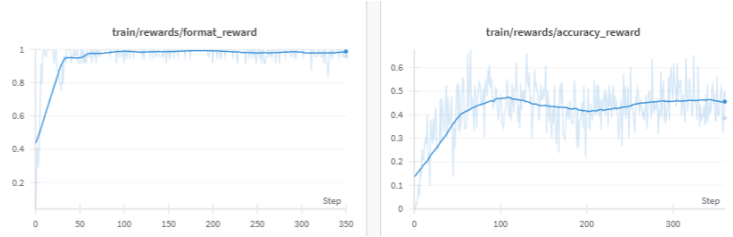


Figure 3: Training curves for the RL fine-tuning phase.

4.3 Results

4.3.1 View Classification Performance

After the initial fine-tuning step, the VLM demonstrated the ability to distinguish between different views based on visual features 4.3.1. The model was able to classify the views with a measurable degree of accuracy, indicating that it has learned some relevant visual differences associated with each view. However, the performance in terms of accuracy was not as high as expected, suggesting that there is room for improvement. One potential reason for this moderate performance could be the way the cross-entropy loss function is computed. Specifically, the cross-entropy for the class token is calculated over the entire vocabulary, rather than focusing solely on the 12 view class labels. This could dilute the model’s attention on the relevant class tokens, leading to suboptimal performance. To address this, a promising approach would be to modify the loss function to compute the softmax only over the 12 relevant class tokens, rather than the entire vocabulary. This adjustment would likely improve the model’s ability to distinguish between the specific classes and could lead to better performance in subsequent fine-tuning stages.

Accuracy	F1 Score	Precision	Recall
0.63	0.58	0.70	0.59

Table 1: View Classification Performance Metrics After Fine-Tuning

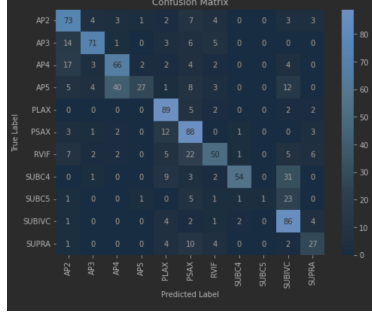


Figure 4: Confusion Matrix for the view classification step.

4.3.2 RL Fine-Tuning Results

For analyzing the results of reasoning performance improvement after RL fine-tuning, we will present both qualitative and quantitative assessments.

First, a qualitative comparison before and after RL fine-tuning 6 clearly illustrates how the model’s reasoning capabilities evolve. Initially, the VLM struggles to generate reasonable justifications for its view classification decisions. Despite being given the relevant "clues" in the prompt, it fails to provide an explanation that aligns with the correct clinical understanding. However, after fine-tuning with the reward model, the VLM begins to integrate some form of justification in its output. For instance, the model now includes anatomical and clinical observations that relate to the predicted view, but they may not always be entirely accurate. The reasoning output will be marked with green and red colors to denote correct and incorrect statements, respectively.

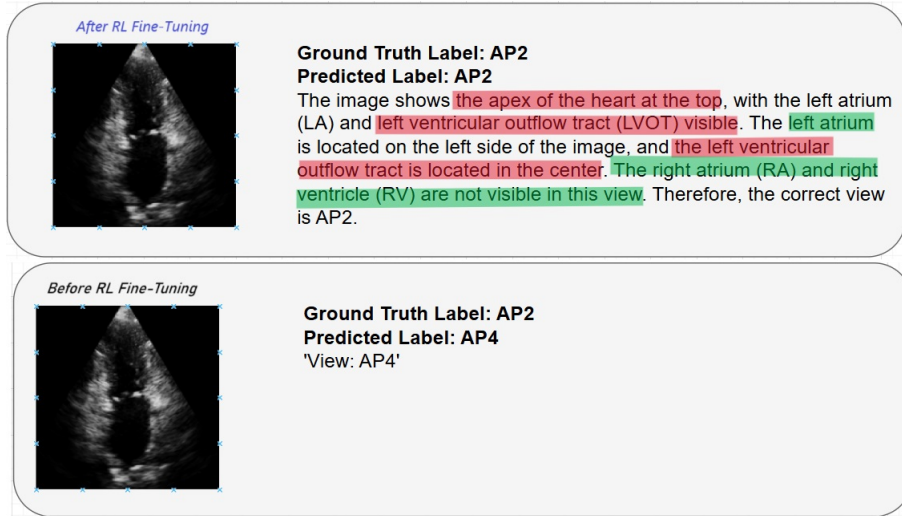


Figure 5: Qualitative comparison of reasoning performance before and after RL fine-tuning. The upper part illustrates the reasoning generated after fine-tuning, where correct statements are highlighted in green and incorrect ones in red. The lower part shows the reasoning generated before fine-tuning, which lacks clear justifications

To further assess the effectiveness of this fine-tuning approach, we evaluated the reasoning on a set of 10 test samples using GPT-4. The evaluation criteria were based on clinical guidelines, with GPT-4 providing a score between 0 and 10 for the quality of reasoning in the generated outputs. The results, as summarized in the table, demonstrate a marked improvement in the reasoning capability of the VLM following fine-tuning. Although the generated reasoning is not yet perfect, these findings indicate that the model’s ability to provide justifications has significantly improved, showcasing the promise of our approach. With further fine-tuning, these results have the potential to become more accurate and clinically relevant.

This analysis demonstrates that while our method is still in its early stages, the improvements in reasoning provide a solid foundation for future enhancements, highlighting the value of incorporating structured feedback into model training.

Before RL Fine-Tuning	After RL Fine-Tuning
2.2	5.6

Table 2: GPT-4 evaluation scores for reasoning performance before and after RL fine-tuning. The table shows an improvement in the evaluation score, indicating that RL fine-tuning has positively influenced the model’s ability to justify its predictions based on clinical guidelines.

5 Conclusion

In this paper, we addressed the challenge of improving the reasoning and classification capabilities of Vision-Language Models (VLMs) for echocardiographic view classification. The task involved not only distinguishing between different view classes based on image input but also providing accurate justifications for these classifications based on clinical guidelines. We proposed a framework that combined reinforcement learning with a rule-based reward model, ensuring that the model learned to incorporate clinically relevant justifications into its reasoning. Our results show that the proposed method improved both classification accuracy and reasoning ability, with GPT-4 evaluations indicating a significant increase in reasoning quality post-RL fine-tuning. Despite some remaining challenges, particularly in achieving perfect reasoning accuracy, our approach demonstrates promising results in improving the interpretability and justification of medical predictions.

There are several avenues for future improvements. Firstly, using a more sophisticated reward model, such as GPT-4, could enhance the model’s reasoning further. GPT-4 has shown great potential as a medical evaluator [3] when paired with carefully designed prompts, and adopting it for this task could refine the justifications provided by the VLM. Secondly, exploring more complex VLMs with larger parameter sizes could provide better feature extraction and representation capabilities, thus improving both classification and reasoning performance. Such models could lead to more accurate predictions and more robust reasoning in the context of medical decision-making, as demonstrated in other works.

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6 Supplementary

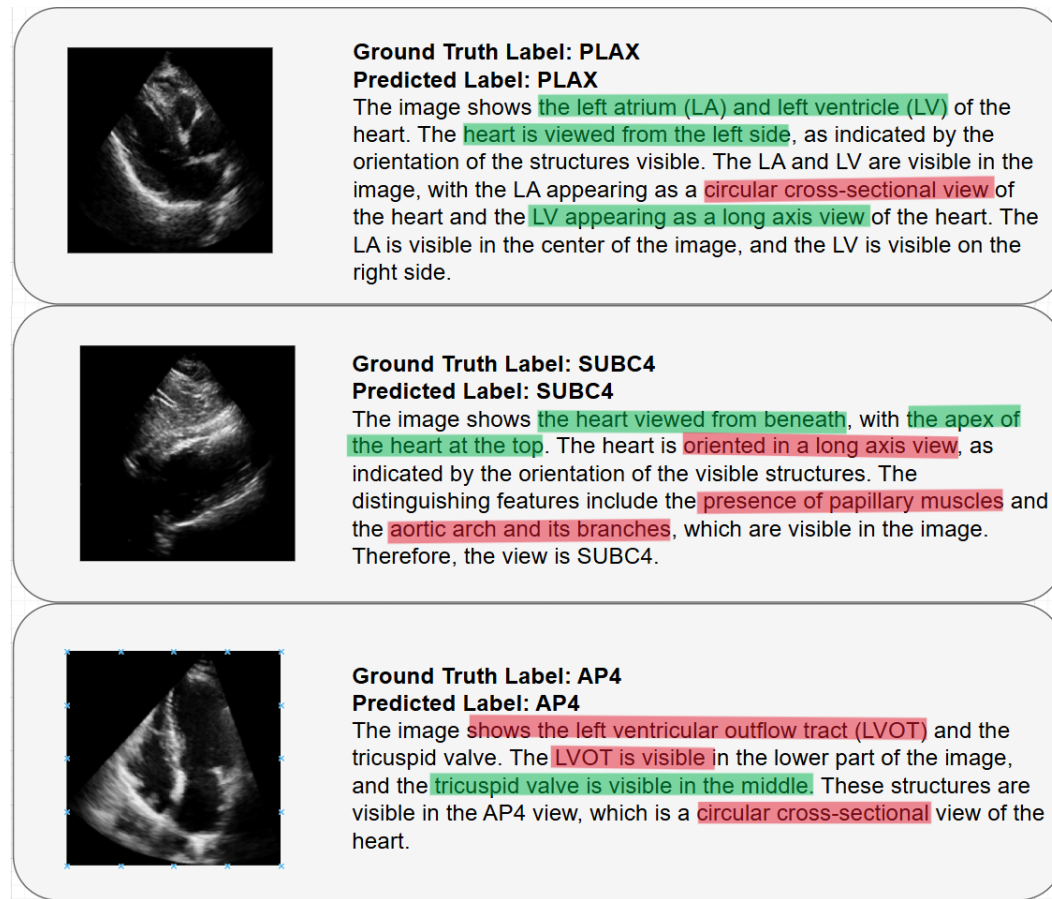


Figure 6: Qualitative results of reasoning performance after RL fine-tuning for different views.